

# The Relational Database Environment

## IEM 4723 Information Systems Design

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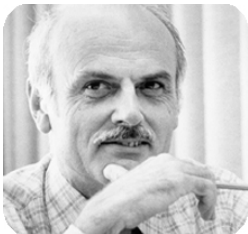
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# The Relational Data Model

# Introduction



Dr. E.F. Codd

- The relational data model was introduced by Dr. E.F. Codd in 1970, building on mathematical set theory and predicate logic.
- It remains the dominant approach for structuring data in the databases used today.

# Relations

| Customer Number | First Name | Last Name | Phone          |
|-----------------|------------|-----------|----------------|
| 0001            | Jane       | Doe       | (555) 555-1111 |
| 0002            | John       | Doe       | (555) 555-2222 |
| 0003            | Jane       | Smith     | (555) 555-3333 |
| 0004            | John       | Smith     | (555) 555-4444 |

Fig. 5.1 A simple customer relation

- The word table is used synonymously with relation in a relational data model. A relation specifies only what will be contained in each column, and the rows of a table are instances of that relation.
- Each column has a name that is unique within its table, and draws its values from a defined domain. Columns have no positional concept, meaning they can be viewed in any order.
- Each row has only one value at the intersection with a column, so there are no multi-valued attributes, and no two rows in a relation may be duplicates. A primary key ensures this uniqueness, and like columns, rows have no positional concept and can be viewed in any order.
- Relations are commonly written using the notation `relation_name(primary_key, non_primary_key_column, ...)`, such as `customer(customer_number, first_name, last_name, phone)`.

# Primary Key

# Concatenated Primary Keys

| Order Number | Item Number | Quantity |
|--------------|-------------|----------|
| 10991        | 0022        | 1        |
| 10991        | 0209        | 2        |
| 10991        | 1001        | 1        |
| 10992        | 0022        | 1        |
| 10992        | 0486        | 1        |
| 10993        | 0209        | 1        |
| 10993        | 1001        | 2        |
| 10994        | 0621        | 1        |

Fig. 5.2 A sample order items table

- A concatenated primary key, also called a composite or compound key, is made up of the smallest number of columns needed to uniquely identify a row, such as `order_number` and `item_number` together in an order items table.
- An all-key relation is one where every column is part of the primary key, such as `subject_catalog(isbn, subject_heading)` or `enrollment(CRN, CWID)`.

# Foreign Keys

| <u>Items</u> |                               |                    |       |
|--------------|-------------------------------|--------------------|-------|
| Item Number  | Title                         | Distributor Number | Price |
| 1001         | Gone with the Wind            | 002                | 39.95 |
| 1002         | Star Wars IV: Special Edition | 002                | 59.95 |
| 1003         | Die Hard                      | 004                | 29.95 |
| 1004         | Bambi                         | 006                | 29.95 |

| <u>Orders</u> |                 |            |
|---------------|-----------------|------------|
| Order Number  | Customer Number | Order Date |
| 11100         | 0012            | 12/18/09   |
| 11101         | 0186            | 12/18/09   |
| 11102         | 0056            | 12/18/09   |

| <u>Order Item</u> |             |          |          |
|-------------------|-------------|----------|----------|
| Order Number      | Item Number | Quantity | Shipped? |
| 11100             | 1001        | 1        | Y        |
| 11100             | 1002        | 1        | Y        |
| 11101             | 1002        | 2        | Y        |
| 11102             | 1002        | 1        | N        |
| 11102             | 1003        | 1        | N        |
| 11102             | 1001        | 1        | N        |

Fig. 5.3 Three relations from the Antique Opticals database

- A foreign key is a primary key migrated from a related table into another table, in order to capture the relationship between them. A database relationship is indicated by matching data between a primary key and a foreign key.

# Referential Integrity

- The referential integrity constraint states that every non-null foreign key value must match an existing primary key value.
- This is the most important constraint in a relational database, since it ensures consistency of the cross-references among tables, and is stored in the database and automatically enforced by the DBMS.
- A foreign key can also reference the primary key of its own table, such as an `employee(employee_ID, first_name, last_name, department, manager_ID)` relation, where `manager_ID` refers to another `employee_ID` in the same table.

# Databases and Data

# Defining a Database

- Data are an essential part of any business today, and are needed to support both day-to-day operations and longer-term decision making.
- Most businesses now store their data in computerized databases.
- Before databases existed, data were organized as simple lists stored in computer files, generally one list per physical file, and managed by list or file management software.
- To be considered a database, a system needs to store not only the raw data, but also information about the relationships among that data, such as the relationships among customers, orders, and products.
- Specialized software called a database management system, or DBMS, is needed to create and manipulate a database. A user or program requests data, and the DBMS maps that request to locations in physical data storage.

# Types of Data

- Text data is a series of characters whose meaning is not important to the system storing it.
- Numeric data includes integers and real numbers.
- Predefined data items are numerical or alphabetical items whose meaning and format are specified explicitly, such as a social security number, account number, transaction date, or course number, and are common in transaction-oriented systems.
- Databases can also store images, audio, and video.

# Data Hierarchy

The diagram shows a table with 8 columns and 5 rows. Annotations include:

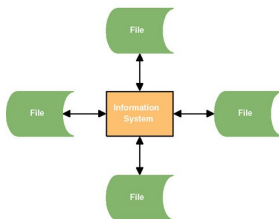
- A callout bubble labeled "the key" pointing to the "Social security number" header.
- A callout bubble labeled "Example of a field" pointing to the "Last name" header.
- A callout bubble labeled "Example of a record" pointing to the entire second row (434-98-8832, Chang, Brenda, 87 Palm Ave., Oakdale, MA, 02143, 09/30/80).

| Social security number | Last name | First name | Street address | City      | State | ZIP code | Date of birth |
|------------------------|-----------|------------|----------------|-----------|-------|----------|---------------|
| 044-34-5542            | Bates     | Alvin      | 243 Third St.  | Middleton | MA    | 02137    | 05/07/78      |
| 434-98-8832            | Chang     | Brenda     | 87 Palm Ave.   | Oakdale   | MA    | 02143    | 09/30/80      |
| 888-23-9038            | Schmidt   | Dieter     | 663 Cress Way  | Cresston  | MA    | 02184    | 12/17/79      |
| 334-59-3087            | Toliver   | Gail       | 743 First St.  | Middleton | MA    | 02137    | 07/02/78      |

- A file is a set of related records.
- A record is a set of data elements that relate to a particular object or activity.
- A data element is the smallest unit of data that has a predefined meaning.

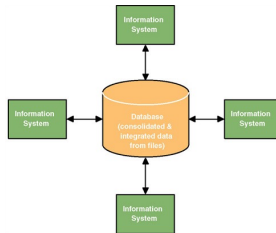
Source: Alter (2002)

# Conventional Files



- A file is a collection of similar records, generally linked to a specific application program, such as an accounting application that keeps track of students' fees and payments.
- The structure of a file is embedded in its application program, and files are unrelated to each other except through that program's code, so data storage ends up built around the applications that use the files.

# The Database



- A database is a collection of interrelated files, forming a single repository of data accessed by various users and applications.
- Records in one file, or table, are physically related to records in another, and applications are built around the integrated database rather than around individual files.
- A database system contains the database itself, along with a complete description of the database called metadata, giving program-data independence, the ability to change the data structure without changing the application program.

# Information Systems and DBMS Software

# Information Systems and Databases

- Online transaction processing, or OLTP, systems handle day-to-day operations such as sales, accounting, manufacturing, and human resources, and form the basis of information processing in most organizations. Data in an OLTP system are dynamic and change frequently.
- Online analytical processing, or OLAP, systems support analysis of organizational performance, high-level decision making, and strategic planning. Data are extracted from OLTP systems, reformatted as necessary, and loaded into OLAP systems.
- Relational databases are mainly used in OLTP systems, store highly structured data, and make up the bulk of the databases used today. Some OLAP systems use relational databases as well, though most require different database structures.

# Database Software: DBMSs

- Enterprise-strength commercial DBMSs, such as Oracle, DB2, and Microsoft SQL Server, are large and expensive pieces of software that require significant training and expertise, and implementing one requires the logical structure of the database to already be defined.
- A DBMS creates the physical structure of the database, provides ways to enter, modify, and delete data, and provides ways to retrieve data and process user-formulated queries.
- A DBMS can also perform basic calculations, and provides methods to restrict access to data, such as password protection and access privileges.

